Early-warning system for bearing faults: a note for plant engineers

What it does. It listens to a motor's vibration and learns what a healthy bearing looks like. When the vibration stops looking healthy, it raises an alert.

What it never saw. It was taught on healthy data only. It was never shown a broken bearing, so it can also react to fault types it has not met.

How it did on our test. On laboratory test recordings it flagged all 18 faulty test recordings (inner race, ball and outer race, at motor loads of 2 and 3 HP). The first alert came within one reading, about 0.09 seconds. It raised no false alerts in the 27.7 seconds of healthy test data we held back.

The honest limit. Our healthy test data lasted only 27.7 seconds, so we cannot promise how many false alerts per day you will see. These were also strong, laboratory-made faults; real wear grows slowly and may be harder to catch. A simple "is the vibration much stronger than usual" rule also worked almost as well on this test, so the test was easy.

The alert level. A missed fault costs about Rs 4 lakh and a false alert about Rs 2,000, so we lean towards catching faults. We recommend alerting when the pattern error is roughly three times its usual healthy level (setting A in the report). It caught every faulty reading in our test and gave no false alert on healthy readings. A higher setting (B) saved nothing in our test and let some faulty readings through.

Before you rely on it. Run it in watch-only mode on your own machine for a few weeks and count the alerts, then set the alert level using your own data. Do not let it stop a machine automatically until then.

What to do on an alert. Check bearing temperature and noise; if both look normal, inspect at the next planned stop.

When to retrain. After a bearing or motor change, or a big change in load or speed.